

TRISTIN WEBB

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EDUCATION

New Mexico State University — Las Cruces, NM

Expected May 2028

Bachelor of Science in *Electrical Engineering*

GPA: 3.72

- Concentration: *AI, Machine Learning, & Data Science*
- Minor in *Computer Engineering & Software Development*

San Juan College — Farmington, NM

May 2025

Associate of Science in *Computer Science*

GPA: 3.96

EXPERIENCE

Undergraduate Research Assistant – WASI Lab, New Mexico State University

Aug 2026 – Present

- Researched and evaluated existing visualization tools and packages for interpreting machine learning models, informing the design approach for an explainable AI (XAI) visualization pipeline.
- Built a directed graph visualization of a reinforcement learning model in Python, applying graph-based data structures to represent state-action relationships and improve model interpretability.
- Studied foundational concepts in reinforcement learning and explainable AI to support visualization design decisions and technical discussions with the research team.

Technology Transfer Intern – Sandia National Laboratories

June 2026 – Aug 2026

- Conducted commercialization and market research for an emerging technology by applying CARAT, Business Model Canvas, Customer Profile, Value Proposition Model, and TAM/SAM/SOM frameworks to assess industry need and adoption potential.
- Expanded industry outreach by sending 200+ LinkedIn connection requests and facilitating 20 customer discovery interviews with professionals across aerospace/defense, industrial manufacturing, and related sectors.
- Identified barriers to entry, customer pain points, and market fit considerations through primary research to inform the technology transfer and commercialization roadmap.

PROJECTS

Wikipedia Flower Dataset: Scraping, OOP Modeling & Visualization | *Python, BeautifulSoup, Pandas, Matplotlib, Seaborn, Regex*

Nov 2025

- Built a Python scraper (Beautiful Soup, Requests, rate-limiting) to collect and export 1,000+ Wikipedia flower records to JSON.
- Cleaned data with Pandas/regex and designed an OOP model using Flower as the base class, and Wildflower as the subclass (inheritance, polymorphism), and a separate Garden class with search/filter methods.
- Visualized color and habitat distributions with annotated bar and pie charts using custom palettes and percentage labels.

2D Platformer Game | *Java, Object-oriented programming, Eclipse IDE*

Nov 2024

- Developed a 2D platformer game in Java using Eclipse, featuring a custom popup game window, keyboard-based directional controls, sprite animation, and collision detection.
- Implemented core gameplay mechanics including player movement, object interaction, and basic physics-based collisions.
- Designed and tested game logic to support smooth animation timing and responsive controls.

Obstacle-Avoiding Autonomous Car Robot | *Arduino, Embedded Systems, C/C++, Arduino IDE*

Nov 2025

- Designed and implemented an autonomous obstacle-avoiding robot car using an Arduino microcontroller, ultrasonic sensor, and servo motors.
- Programmed control logic in Arduino IDE to detect obstacles, navigate the environment, and adjust movement in real time.
- Tested, debugged, and demonstrated the final prototype to verify reliable operation and responsive obstacle avoidance.

LEADERSHIP

Officer – Institute of Electrical and Electronics Engineers

Aug 2025 – Present

- Facilitated and supported Open Project Space (OPS) 2.0, coordinating student project activities, collaboration, and event logistics.

Tutor/Mentor – New Mexico State University

Aug 2025 – Present

- Tutored students in Computer Science, Electrical Engineering, Mathematics, Physics, and related subjects.
- Mentored students in developing effective study skills, problem-solving strategies, and academic confidence.

SKILLS

Technical: C, C++, C#, Python, Java, GitHub, VHDL, SQL, Excel, PivotTables, Tableau, MATLAB, Visual Studio, VS Code, Arduino IDE

Analytical: Data Analysis, Data Visualization, Market Research

AWARDS

S-STEM Scholar; Math Excellence Award; Student Leadership Award; Crimson Scholar; Phi Theta Kappa Honor Society; SWASAP Award